

Timeshift Media Coding Exercise

Thank you so much for your interest in Timeshift Media. We're excited to have the opportunity to interview you. The following questions are something for you to work on in your own time. We expect you to spend about 2 hours working on this. If you spend more time, we won't be giving bonus points for more work. If you have any questions while preparing your answers, please don't hesitate to reach out to adam.conner@timeshiftmedia.com with any questions.

Use React Native and typescript as the primary language for this exercise, and be prepared to speak to your solution. During the interview we'll review this together and do some pair programming using your solution as a starting point.

The questions below are based on the game Wordle. You may find lots of examples of other Wordle implementations online. Use what you decide, or create your own. The goal is to speak to what you've implemented and demonstrate your knowledge and skills. If you're not familiar, the game and rules are available here: <https://www.nytimes.com/games/wordle/index.html>

Questions:

1. Create a data type which represents the Wordle game board. This structure should be able to describe the current state of the game, and if complete if the game was won or lost.
2. Create a function which accepts the data type in the first question and the word to test for and returns the game state. Appropriate states can represent:
 - a. Correct guess, game complete.
 - b. Incorrect guess, number of remaining guesses, letters used, correct and incorrect letters and positions.
 - c. Incorrect guess, game over
3. Create at least positive and at least one negative test case which validates the function in question two.
4. (If time allows) Create or start creating a front end form which represents the questions above.